



Citation (Version of Record):

Godínez Castellanos, J. L., Oliver, T. A. A., Larsen, K. A., Brahms, C., Travers, J. C., Forbes, R., Bain, M., & Bradforth, S. E. (2026). Ultrafast solution-phase photochemistry revealed by multi-octave continuum probe. *The Journal of Physical Chemistry Letters*, 17(9), 2634–2643. <https://doi.org/10.1021/acs.jpcllett.5c04072>

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[10.1021/acs.jpcllett.5c04072](https://doi.org/10.1021/acs.jpcllett.5c04072)

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Ultrafast Solution-Phase Photochemistry Revealed by Multi-Octave Continuum Probe

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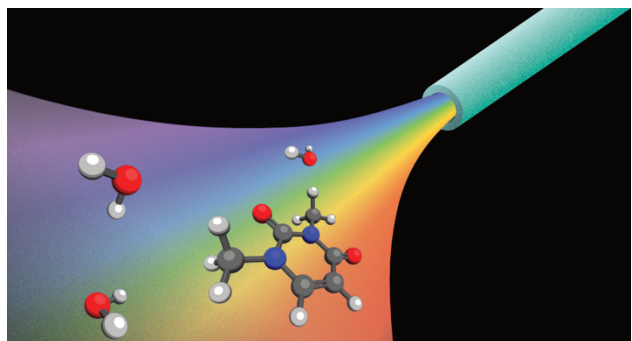
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Abstract

A transient absorption (TA) spectroscopy experiment using a multioctave supercontinuum probe simultaneously covering the visible, near and deep ultraviolet is developed and employed to reveal new photochemical phenomena for two aqueous phase systems. Although this spectral region contains key absorptions associated with amino acids and DNA building blocks, supercontinua extending to 5.5 eV with sufficient bandwidth and photon-flux have been difficult to achieve for solution phase spectroscopy. This technical challenge is surmounted by utilizing gas-filled hollow-capillary fibers to generate bright ultrafast super-broadband soliton probes using both Ti:Sapphire and Yb:KGW amplifiers. Here, we demonstrate soliton-probe continuum generation spanning 2.1–5.5 eV for liquid-phase TA spectroscopy. We apply this innovation to the continuous two-photon absorption of liquid water, revealing new polarization information on pre-resonant states, and to the UV photohydration reaction of RNA derivative, 1,3-dimethyluracil (DMU). In DMU, we directly detect a <2% channel, demonstrating that the photohydration reaction occurs via a twisted ground-state intermediate.

TOC Graphic



The advent of femtosecond time-resolved spectroscopy in the 1980s ignited extensive research focused on understanding the photochemistry and photodynamics of excited molecules. In its infancy, femtosecond spectroscopy was limited to single-wavelength pump and probe pulses that hindered insights into spectral evolution following excitation, and consequently, a comprehensive understanding of the excited state dynamics. These probe pulses were either generated as harmonics of the ultrafast laser fundamental, in optical parametric amplifiers (OPAs) or selected with bandpass filters from continua generated in bulk media.¹ The development of white light supercontinuum with broader bandwidth² in the late 1980s and early 1990s propelled several research groups to implement multichannel detectors for spectrally resolved probing in femtosecond laser spectroscopy.^{3–5} These technical advances laid the foundations for contemporary versions of transient absorption (TA) spectroscopy—now a familiar tool because of the rich insights it delivers in many disciplines spanning the life^{6,7}, energy^{8,9} and materials sciences.^{10,11}

In present-day experiments, supercontinua for TA experiments are generated in different bulk media such as CaF₂, sapphire, liquid water or YAG, pumped either using the fundamental or harmonics of ultrafast laser amplifiers,^{12,13} CaF₂ has become one of the bulk media of choice as it can be used to generate a usable supercontinuum that spans the near-ultraviolet (UV) to all visible wavelengths 320 – 750 nm when pumped with the fundamental of a Ti:Sapphire laser.¹³ Extension into the UV is possible by pumping CaF₂ at 400 nm or 266 nm but the bandwidth obtained is limited.^{12,14} Alternatively, UV continua can be obtained through more sophisticated methods such as achromatic frequency doubling in nonlinear optical crystals.^{15,16} The wavelength regions that these different approaches can span are illustrated in Figure 1 (a).

Generation of supercontinua in bulk media has two major limitations: the bandwidth extends over less than one octave, and the photon flux in the UV - where detector sensitivity is already low - is insufficient to achieve a good signal to noise ratio. The latter poses a significant challenge, as this region is where critical absorption bands are found for the photoproducts of small organic molecules. Specific examples include products comprised of a single aromatic or partially conjugated ring, olefins, or only aliphatic hydrocarbons, which are both common functional groups of many biomolecules.¹⁷⁻¹⁹ In addition, for molecules with absorption onsets in the deep UV, a ultraviolet probe enables measurement of the bleach recovery and stimulated emission features which are key in confidently assigning the kinetics. Another problem where broad spectral coverage is particularly valuable is continuous coherent two-photon absorption spectroscopy using a super-continuum probe which provides complementary electronic spectroscopy information to routine one-photon absorption (UV-vis). Features that occur at equivalent transition wavenumber to one-photon absorption (1PA) vacuum ultraviolet wavelengths are readily obtainable and allow characterization of high lying molecular electronic states.^{20,21} Currently, to collect time-resolved or two-photon absorption (2PA) spectra over a very broad spectral region, separate sequential measurements with narrower probe bandwidths must be acquired followed by laborious data stitching.^{22,23} This approach is challenging to implement, and thus experiments reported in the literature are largely constrained to narrower and more limited spectral windows, for example 320 – 750 nm of a CaF₂ super-continuum being the most popular. Hence, there is a compelling need to innovate new continuum sources extending beyond one octave. Taking advantage of emerging advances in nonlinear optics in gas-filled fibers, covering wavelengths from the NIR into the vacuum UV (VUV) with equivalent spectral density throughout the spectrum is now possible.^{24,25}

Soliton continuum sources can fulfill this technical demand by providing a seamless multi-octave probe that allows us to capture dynamic broadband spectral information in a *single* transient absorption measurement. In the recent past, gas-filled photonic-crystal fibers have been used to generate multi-octave supercontinua.^{26,27} However, recent advances lead to the realization that the same feat can be achieved using gas-filled hollow capillary fibers (HCFs) using few cycle pulse compression—a crucial step as HCFs allow for the use of much higher energies because of their lower damage threshold.²⁸ These HCF sources have been proven to generate over two octaves of spectral bandwidth from the deep UV (120 nm) to the NIR (>1000 nm).²⁸ However, only a single report has appeared where HCFs have been used as a continuum probe for condensed phase time-resolved spectroscopy – and the application was transient reflectance measurements for a solid state system, with a very large transient signal equivalent to around ~190 mOD.²⁵ Here we show the successful implementation of multi-octave supercontinua driven by *both* Ti:Sapphire and Yb:KGW ultrafast amplifier lasers and applied to transient absorption for more demanding molecular applications where the transient signal is at least an order of magnitude smaller than in solid state transient reflectance: solution-phase photochemistry and coherent 2PA spectroscopy of a bulk liquid. We demonstrate the power of this approach through two examples that uncover new solution phase phenomena: (1) the two-photon absorption spectrum of liquid water reveals the isolated molecule parentage²⁰ of its collective optical conduction band and (2) by directly observing a small but important photochemical product channel, photohydrate formation, we learn about the mechanism of UV photodamage in the RNA base derivative, 1,3-dimethyluracil. The latter example showcases the high sensitivity to solution-phase transient species when implemented in absorption.

Solitons were generated in gas-filled HCFs using two different ultrafast lasers, demonstrating that multi-octave supercontinua are not restricted to a specific laser platform. Using Ti:Sapphire (800 nm, 50 fs) and Yb:KGW (1030 nm, 200 fs) laser systems, 1 mJ at 1 kHz was used to drive soliton generation. In both instances, the optical design and sequence of HCFs were informed by non-linear optical dynamical simulations in the Luna.jl software package.²⁹ Earlier work indicated that soliton generation is best facilitated by driving laser pulses with near to 10 fs duration, making it optimal to first compress the fundamental output of the ultrafast amplified laser source.²⁸ However, pulses this short are not mandatory.^{28,30} In our work, pulse compression was accomplished with a single compression stage using a single HCF for the experiment using a Ti:Sapphire amplifier, and two sequential compression stages with the Yb-KGW laser due to the four-times longer initial fundamental laser pulse duration.³¹

The ideal continuum for use in pump-probe spectroscopy has a multi-octave spectral bandwidth and uniform energy density. The full experimental details of our implementation of HCFs for soliton generation are given in the supporting information (SI) – but the key ideas and considerations required for a robust source for solution phase transient absorption experiments are covered here. In a soliton the flattest continuum is found at the point where the soliton fission occurs. Therefore, our setup was optimized for fission to occur at, or very close to, the fiber exit.^{27,32} Under these conditions, the soliton reaches its maximum bandwidth, and if it were to propagate through the fiber further, the soliton would undergo fission thereby dramatically decreasing the spectral bandwidth—see numerical simulations in Figures S.3 and S.4. These design characteristics were considered in the implementation of the second HCF soliton stage (150 cm length, inner diameter = 150 μ m, 1.9 bar of Neon gas). A pair of fused-silica wedges were used

to alter the pulse duration launched into the soliton generation stage, and to tune the output pulse characteristics.

A typical soliton spectrum used for transient absorption measurements is displayed in Figure 1(b) and is characterized by an excellent spectral range from < 200 nm down to 1000 nm and just one order of magnitude drop in intensity from the fundamental to the shortest wavelength in the DUV—far exceeding the 3-4 orders of magnitude drop in intensity observed in bulk media.¹³ Note that this is a lower limit for the immediate soliton bandwidth out of the HCF, as the spectrometer has a finite sensitivity range, and any VUV wavelengths generated will be absorbed by ambient air. Crucially, the intensity of the driving fundamental at 800 nm is only one order of magnitude more intense than the region between 200–300 nm. The resulting laser spectrum is far superior to supercontinua generated in bulk materials such as sapphire where the intensity plateau of visible wavelengths is often >2 orders³³ lower and it is not possible to generate such a broad bandwidth, especially one extending this far into the deep UV.^{12,13}

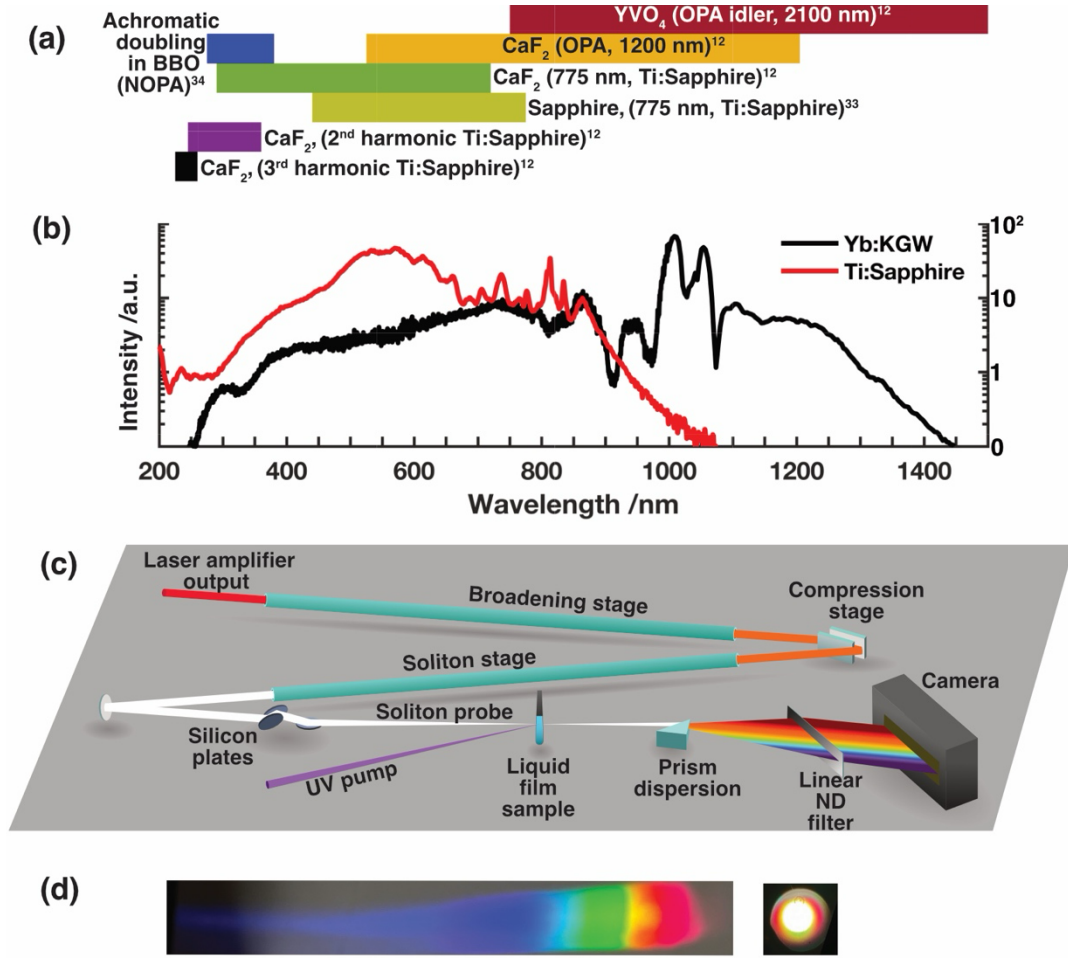


Figure 1 Multi-octave soliton continua applied here for solution phase transient absorption. The exceptional soliton supercontinua bandwidth generated in hollow capillary fibers is compared with supercontinua from solid state materials. (a) Piece-wise continua used in different literature experiments for liquid-phase transient absorption, from the deep UV to the NIR.^{12,33,34} (b) Spectra for the soliton supercontinua generated using Ti:Sapphire and Yb:KGW laser systems and implemented for a transient absorption experiment in the liquid phase. The pulse energy obtained for the soliton spectra shown here are 20 μ J for Ti:Sapphire system and 60 μ J for Yb:KGW. The soliton supercontinuum surpasses both in bandwidth and the overall pulse energy those generated in bulk media (typically only a few μ J).¹³ (c) Simplified representation of key components of the soliton-probed transient absorption setup implemented here. For a detailed description of every component see SI. (d) Dispersed spectrum and spatial mode of the soliton supercontinuum.

The corresponding soliton spectrum generated with the Yb:KGW laser spans from 300–1400 nm (overlaid in Figure 1(b)), producing almost three octaves of laser bandwidth. The longer fundamental Yb:KGW laser wavelength biases the center of the supercontinuum towards the near-IR as opposed to the deep UV. The deep-UV is a particularly important region to have access for

probing solution phase photochemistry. Red-shifting the fundamental driving field for soliton generation was also demonstrated by Brahms *et al.*, where the 1750 nm output generated from an optical parametric amplifier yielded a supercontinuum spanning 220–2250 nm, as applied to solid-state transient reflectance spectroscopy.²⁵ Despite differences in their generation and spectral content, both continua were proven as effective ultrafast probes as shown by implementation in the main text (Ti:Sapphire) and their comparison in the SI (Ti:Sapphire and Yb:KGW).

As shown in Figure 1(c), the collimated output of our Ti:Sapphire soliton supercontinuum was spectrally filtered by two silicon plates to selectively attenuate the near-IR and visible portions of the spectrum, without impacting the UV wavelength range.³⁵ The polarization purity of the probe pulse was further improved by the reflection off a third silicon plate. While non-linear processes in HCFs generally preserve the input polarization of the driving pulse any slight deformation in the capillary core can reduce the polarization contrast.^{32,36}

To implement solution-phase transient absorption, the resulting multi-octave probe pulse measurements was focused into a flowing gravity-driven wire-guided jet (100 μm pathlength),³⁷ with an off-axis parabolic (OAP) mirror (EFL = 203.2 mm). The 267 nm pump pulse, shown with less detail in Figure 1(c), was derived from a portion of the 800 nm fundamental mixed with its second harmonic at 400 nm by type I sum frequency generation (type-II BBO, 150 μm) and focused into the liquid sample with a CaF_2 lens ($f = 25$ cm) to achieve a pump-spot size between 100–170 μm . The 267 nm pulse was compressed prior to the sample by a pair of Gires-Tournois negative dispersion multilayer mirrors,³⁸ and set to magic angle with respect to the soliton probe by a zero-order $\lambda/2$ waveplate.

After the sample, the soliton probe beam was collimated with a second identical OAP mirror and coupled into a home-built spectrograph. A CaF_2 prism-based spectrometer overcomes the unavoidable folding of higher diffraction orders that would occur for a multi-octave probe used with a grating spectrometer and accomplishes more evenly dispersing the probe approximately by frequency (rather than wavelength) onto a line diode array. Full details of the optical elements and calibration of the spectrometer are given in the SI, but key elements include a 1 kHz 16-bit detector (EB Stresing) with a 1024-element back-thinned CCD image area sensor (Hamamatsu, S7034-1007S). Critically, the thin quartz window on the front of the CCD detector has $> 90\%$ transmission over the spectral response of the sensor (200–950 nm), and the sensor has a $> 40\%$ quantum efficiency.

The first application for a soliton probe in liquid phase transient absorption spectroscopy is demonstrated measuring a continuous and broadband two-photon absorption (2PA) spectrum of liquid water itself. Continuous 2PA provides access to important spectroscopic information about the electronic structure of condensed phase molecules, for example the fate of Rydberg states in liquids³⁹ by comparing to gas phase electronic spectra,⁴⁰ and by taking advantage of the different one- and two-photon transition symmetry selection rules as well as the extended spectral range that can be covered.²⁰ An understanding of the spectrum of excited states of condensed water is critical information in informing radiation, X-ray and strong field induced ionization of water.^{41–43} Such selection rules are most powerful for high-symmetry molecules but are important even for a C_{2v} system like water.^{44–47} Further, the relative polarization of the two photons can help to elucidate the symmetry of the excited states and help disentangle broad overlapping spectral transitions often seen in the liquid phase, because of differing polarization ratios. With the combination of a 267 nm (4.6 eV) pump and our soliton multi-octave supercontinuum probe (2.2–

5.5 eV), we can readily access the necessary high combined photon energy spectral region, 6.8–10.2 eV, corresponding to the band-gap of water. This is difficult to access at the one-photon level because of the VUV wavelengths required.^{40,48–50} In prior measurements from our lab, with conventional supercontinua generated in CaF₂ when combined with a 267 nm pump pulse, we could only capture a relatively narrow portion of the valence 2PA spectrum between 7.4–8.7 eV.^{20,51}

Conventionally, TA spectra are displayed as a function of probe wavelength, but due to the significantly extended bandwidth our soliton probe affords, and the choice to disperse the supercontinuum in a prism spectrograph, the data presented here are instead displayed as probe energy in electron volts. The 4.6 eV pumped, soliton probed 2PA data for water at the parallel configuration is shown in Figure 2 (a), collected with a Ti:Sapphire laser system. The water 2PA signal intensity is greatest in the > 4 eV probe region previously inaccessible with a regular CaF₂ probe (2.0–3.8 eV) and the same pump excitation energy. Notably, the chirp is far greater across the entire bandwidth of the soliton probe compared to the CaF₂ supercontinuum due to inevitably larger differences in group velocity delays in transporting the different spectral components. Further, the non-resonant water 2PA response provides a facile method to characterize the system instrument response. The pulse FWHM varies between 70–80 fs in the visible region and 160 fs in the deep UV.

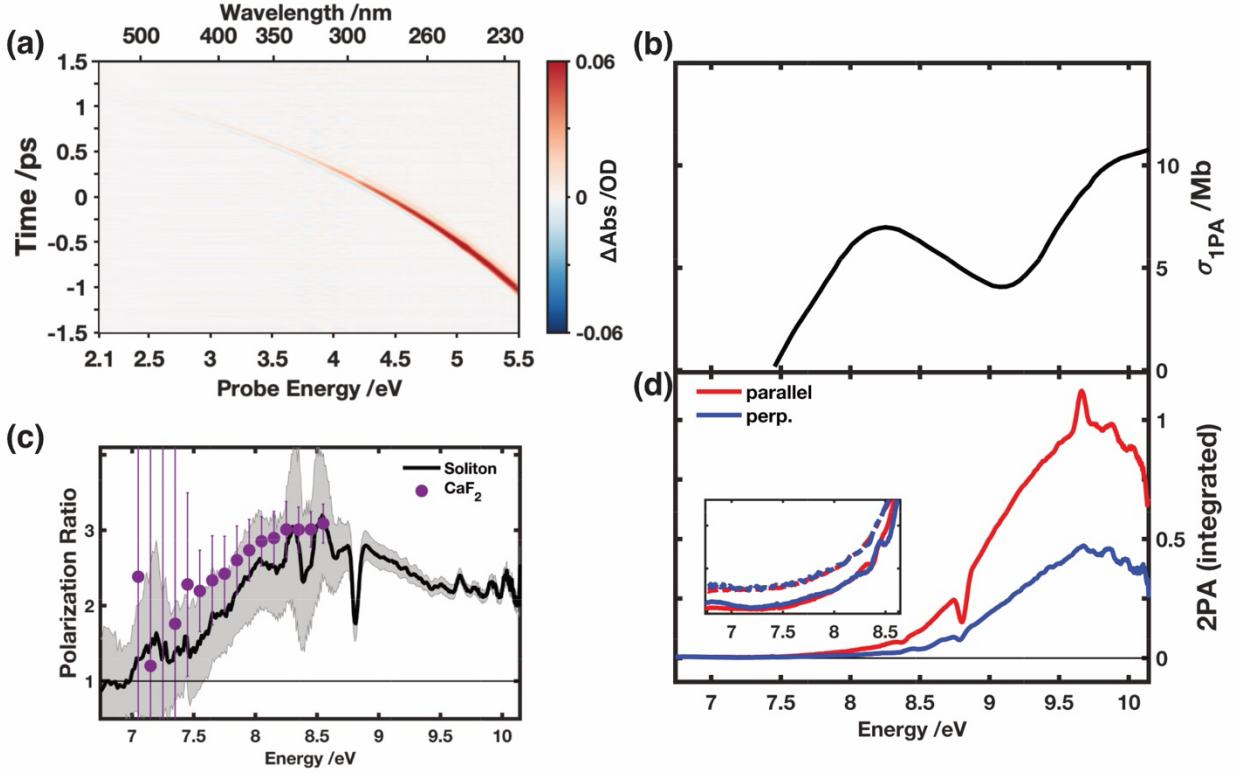


Figure 2. Two-photon absorption (2PA) of water using a 4.6 eV pump and soliton probe in a 100 μm thick liquid jet. Note that the horizontal axes in panels (c, d) are presented using the 1-photon equivalent probe energy. (a) Contour map of the transient non-resonant 2PA signal. (b, d) 1PA⁴⁹ and 2PA spectra of liquid water. Parallel (red) and perpendicular (blue) 2PA spectra of water shows the spectral differences with different polarization combinations. Strong stimulated Raman scattering bands from the O–H stretch are also observed superimposed on the 2PA spectra as gain and loss features near 8.80 and 9.67 eV. The inset shows a close-up of the 2PA spectrum at energies < 8.5 eV; reproduced spectra²⁰ collected using a 4.6 eV pump and a supercontinuum generated in CaF_2 (dashed) is shown with a vertical offset to compare to the spectra collected using the soliton probe (solid), these lines are normalized to 8 eV. (c) The polarization ratio (black, solid line) $\sigma_{\text{para}}/\sigma_{\text{perp}}$ of the 2PA spectrum; the water Raman bands account for the sharp features in the polarization ratio spectrum, and the gain and loss features have polarization contributions of differing sign. The shaded regions are $\pm$ one standard deviation from the average. The polarization ratio obtained using a CaF_2 probe is shown as violet circles with error bars corresponding to $\pm$ one standard deviation.²⁰

Figure 2 shows the 2PA spectrum of water acquired with parallel and perpendicular pump/probe polarizations in panel (d) and the derived polarization ratio, $\sigma_{\text{para}}/\sigma_{\text{perp}}$ in panel (c).

There is a striking contrast between the solution phase water one-photon absorption from the

literature^{49,50} (Figure 2 (b)) and the 2PA (Figure 2 (d)) spectrum: the $\tilde{A}$ band at 8.3 eV associated with the 1^1B_1 state is almost entirely missing in the 2-photon spectrum. Its absence can be explained by the largely oxygen-centered ($3s/4a_1 \leftarrow 1b_1$) transition, which is Laporte forbidden in two-photon absorption. The band peaking at 9.8 eV in the water 2PA spectrum corresponds to multiple overlapping electronic transitions and was only previously observable using a 6.2 eV pump with a conventional supercontinuum probe.²⁰ We note that the 2PA spectrum presented here with a 4.6 eV pump shows a different polarization ratio at > 8.5 eV than prior measurements conducted with 6.2 eV. These differences arise because 6.2 eV excitation is pre-resonant with the first optical absorption band of water and therefore have different 2PA cross-sections, as well as a different polarization tensor,⁴⁵ thus leading to different relative cross sections between the two polarizations. This reinforces the point that new spectroscopic information about the liquid 2PA that can be gained from using a soliton probe and a pump that is fully non-resonant.

The sharp features at 8.80 and 9.67 eV, which are superimposed on top of the broad water 2PA electronic absorption spectrum in Figure 2 (d) correspond to stimulated Raman scattering (SRS) signals. These manifest as Raman loss or gain signals in the water 2PA spectra probed with a soliton continuum, and are only observable due to the availability of a significantly broader bandwidth probe compared to the prior 2PA measurements using a CaF_2 continuum probe.²⁰ The bands correspond to the O–H stretch Raman bands of water at 3250 and 3450 cm^{-1} .⁵² The preferential polarization of the two SRS bands has been observed in prior Raman studies,⁵³ and manifests as sharp features in the water 2PA polarization ratio⁵⁴ at 8.80 and 9.67 eV (Figure 2 (c)).

The polarization ratio associated with the water 2PA in the 8.2–9.5 eV region is consistently above 2.3, growing to a maximum ratio of 3.0, with an estimated standard deviation of 20–30% in this region. On the lower-energy side of the 2PA spectrum, where we can make a

comparison with earlier measurements probed with CaF_2 ,²⁰ the polarization ratio gradually falls to 1.0 with descending total photon energy. The water 2PA polarization ratios derived from soliton and CaF_2 probed measurements are within error, but the polarization ratio obtained with the soliton probe is slightly lower at the lowest total photon energies. The agreement, however, improves with increasing total photon energy, and by 8.5 eV the polarization ratio of the two measurements overlay. This comparison verifies that our supercontinuum is polarized, but the polarization contrast is slightly lower in comparison to a supercontinuum generated in CaF_2 . Overall, the 2PA study of water shows that the soliton supercontinuum is an excellent probe for solution phase TA spectroscopy and has an acceptable polarization purity.

One of our primary ambitions was to develop a spectroscopic probe capable of capturing the ultrafast photoproduct formation of species that have deep-UV absorptions in biological molecules. Here we apply our new experimental apparatus to 1,3-dimethyluracil (DMU), an RNA base derivative and a well-known photochemistry actinometer.^{55,56} Pyrimidines (*e.g.*, uracil, thymine, and DMU) are known to produce two main photoproducts after absorption of UV radiation: a photohydrate and a photodimer.^{57,58} Both photoproducts absorb in the DUV, making their detection by transient absorption challenging.^{58,59} While the quantum yield for photodimerization is comparable among the three pyrimidine bases as aggregates (where bases are stacked making them prone to dimerization⁶⁰), the quantum yield for photohydration of DMU is remarkably two orders of magnitude higher compared to those of the individual canonical RNA and DNA bases and comparable to that of uridine.^{57,58} This high photohydrate product yield makes DMU a prime candidate to understand the competitive photochemistry taking place in canonical DNA and RNA bases, thymine or uracil, and to determine the debated mechanism for pyrimidine photohydration. Two possible mechanisms have been proposed: one involves direct photohydrate

formation on the $^1\pi\pi^*$ excited state^{61,62} while a second involves reaction on the hot ground state,^{57,63,64} with the latter being the currently favored mechanism. The proposed ground state reaction involves a twisted reaction intermediate – structure shown in Figure 4, proposed as the precursor to photohydrate formation.⁶³ To complicate matters further, recent interest^{65,66} in the twisted intermediate emerged from proposals that it may be responsible for the tens-of-picoseconds-lived species detected following electronic relaxation in uracil, thymine, and its derivatives – a species that was formerly attributed to the $^1n\pi^*$ state.⁶⁷

To date, it has been technically challenging to resolve this outstanding problem with ultrafast spectroscopy, as it requires an experiment with femtosecond supercontinuum probe with sufficient bandwidth in the UV to simultaneously track spectroscopic features associated with the photoproduct formation (> 5 eV), the excited state decay of DMU (< 4 eV) into the hot ground state (> 4 eV), and the ground state bleach recovery (4.5–5 eV).^{57,68,69} The lack of a DUV probe has limited the efforts to resolve the ultrafast photochemistry of DMU, with earlier studies unable to resolve earlier than 5 μ s.^{61,67} Recent work^{65,66} has reignited interest in understanding relaxation dynamics and the role of the formation of the twisted intermediate of DMU and other pyrimidines, yet information on photohydrate formation is missing because of the absence of a suitable deep UV probe. As we will show, the power of the soliton supercontinuum for a situation such as DMU is most evident, as we can now time resolve the spectral evolution of all key features simultaneously.

The transient absorption spectra of a ~ 10 mM solution of DMU after 4.64 eV excitation are shown in Figure 3. The false-color contour plot shown in Figure 3 (a, c) illustrates the early time evolution of the DMU transient absorption data; the (negative) stimulated emission (3.5–4 eV) and (positive) excited state absorption (< 3.4 eV) signals associated with the $^1\pi\pi^*$ state both

disappear within on a timescale of 100 fs, or faster, as observed in similar uracil derivatives ($\tau < 100$ fs).^{69,70} These give way to a (positive) photoinduced feature between 3.9–4.3 eV that corresponds to the hot ground state after internal conversion from the $^1\pi\pi^*$ state, as observed for similar pyrimidines.^{69,71}

The spectral characteristics of the vibrationally hot ground state and its thermalization is discussed in greater detail in the SI. In brief, Figure 3(a) shows the characteristic shift of the hot ground state absorption towards higher probe energies that simultaneously fills in the (negative) bleach, as excess vibrational energy is dissipated into the surrounding solvent recreating DMU near the bottom of its ground state potential.⁷¹ With the currently-achieved probe bandwidth we are also able to show that, at early time delays, the (positive) hot ground state absorption also extends to the higher probe energy side (5.0–5.5 eV) of the main ground state bleach, consistent with the temperature-dependent steady-state lineshape (see Figure S11). Figure 3(b) shows the ground state bleach peaks at 4.6–4.7 eV and its recovery from cooling both the DMU vibrational modes and heat dissipation of the local solvent shells is fully complete by 35 ps (Figure 3 (c)) with ~3% intensity remaining at 520 ps, the limit of our temporal window.

The incomplete bleach recovery is of considerable interest as it is due to either photohydrate formation and/or potential branching into the triplet manifold—the latter species is expected to absorb between ~2.0 and 3.5 eV based on prior studies of uracil and its derivatives.^{72–74} A broad absorptive feature peaking at 2.2 eV is in fact observed at longer times (see Figure 3 (b)). Based on its absorption at lower energies, it is attributed to the $^3\pi\pi^*$ species, but the assignment is tentative as the associated absorption cross-section is unknown. As the quantum yield for triplet formation is expected to be an order of magnitude lower than that of photohydrate

formation in DMU,^{57,75} the majority of the bleach remaining at $t = 520$ ps must originate from photoproduct generation.

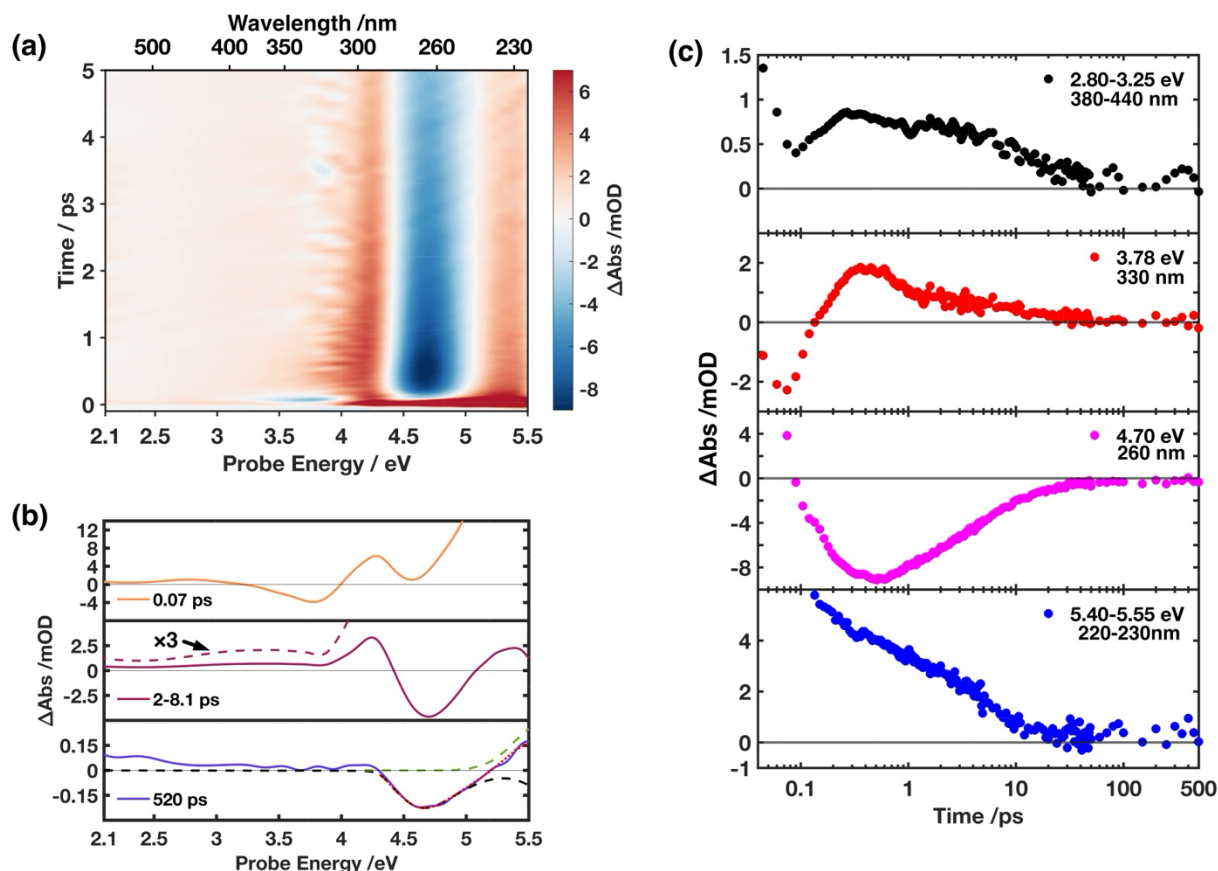


Figure 3 Transient absorption spectra of ~ 10 mM 1,3-dimethyluracil in water after 4.64 eV excitation. (a) Contour map highlighting the first 5 ps of transient evolution. (b) Spectral slices at three time points show the early time signals of the excited state absorbance and stimulated emission (orange) before the appearance of the twisted intermediate (violet, probe energy < 4 eV) and the unrecovered bleach and photohydrate absorption (purple). The ground state spectra of DMU (black dashed line) and the photohydrate⁵⁹ (green dashed line) scaled to the amplitudes needed to reconstruct the TA signal (see SI for details). Notice the large change in absorbance range plotted for each panel. Adjacent delay-binning was used to improve the SNR for the intermediate delays slice, and $200\times$ data averaging for the weak 520 ps signal, to increase the signal-to-noise ratio. In both (a) and (b), to remove spectral modulations between 3.5 – 4.2 eV probe energies an FT filtering scheme has been applied (see SI for details of data workup). A comparison with raw data, as well as description of the filtering is described in the SI. (c) Kinetic traces corresponding to the twisted intermediate (black), stimulated emission hot ground state (red), ground state bleach (magenta), and the hot ground state absorption and the photoproduct (blue).

Figure 3 (c) shows the kinetics associated with the photoinduced absorbance between 5.40-5.55 eV exhibit a fast decay within the first 10 ps, attributed to the cooling of hot ground state DMU molecules. Beyond these time delays, a small photoinduced absorbance is still observed at 520 ps that we attribute to the photohydrate of DMU. The spectral evolution in this region reinforces this assignment. In the second panel of Figure 3(b), the photoinduced absorption in the high-energy probe region at early time delays (< 10 ps) is associated with overlapping ground-state bleach and hot ground-state absorption, exhibiting a plateau between 5.2 and 5.3 eV. As both the second derivative-like spectral structure of the hot ground state and ground state bleach decays away, the spectral maximum of the photoinduced absorption in this region shifts to higher probe energies, revealing the photohydrate which has an associated absorption maximum at > 5.5 eV. The TA spectrum at 520 ps (last panel of Figure 3 (b)) strongly support this view, as the remaining ground state bleach signal matches with the ground state spectrum of DMU up to 4.6 eV, but at probe energies > 4.6 eV the signal rises in intensity and changes sign, with the transient photoinduced absorption agreeing well with the known absorption spectrum of the DMU photohydrate.⁵⁹ This spectral evolution is clearest in a time-resolved simulation (see SI) disentangling the three species that contribute to this spectral region: hot ground state absorption, ground state bleach, and photohydrate. Using carefully derived parameters to match the experimental spectra, we show that there is a negligible contribution from the hot ground state absorption at 520 ps. Critically, this means that the main spectral contributions at this time delay originate from the photohydrate absorption and the associated bleach resulting from the loss of DMU molecules that have reacted with the surrounding water.

Further, reconstruction of the TA spectrum between 4.3–5.5 eV allows for the concentration of photohydrate to be determined, and thus a quantum yield of 0.045 for

photohydrate product to be calculated. This value is in good agreement with the previously reported value of 0.013⁵⁷ - further details of this calculation are given in the SI.

In light of the strong evidence for photohydrate formation at 520 ps, we return to examine the earlier time delays between 2–8.1 ps (middle panel of Figure 3 (b)), we observe three main features: the ground state bleach and hot ground state at probe energies > 4 eV and a broad photo-induced feature at < 4 eV, which peaks at 3.42 eV (dashed line, multiplied $\times 3$ for clarity). This feature appears in a similar region to transients previously reported in uridine by Borrego-Varillas *et al.* (~ 3.60 eV) and calculated for uracil by Obara *et al.* (~ 3.76 eV), which have subsequently been assigned to the absorption of the ground state twisted-intermediate.^{65,66} Furthermore, we note that this feature appears at ~ 100 fs and rapidly disappears over tens of ps (Figure 3 (c)), similar to previously reported for uracil in H₂O and D₂O. For D₂O, the decay associated with the twisted intermediate has been reported by Obara *et al.* ($\tau = 24$ ps) and Mathew *et al.* ($\tau = 31$ ps), whereas for H₂O the reported decay is $\tau = 23$ ps by Obara *et al.*^{65,66} On the basis of similar absorbance and kinetics, we attribute this broad feature at probe energies < 4 eV to the twisted intermediate of DMU. From the first two panels of Figure 3(c) we observe that as the $^1\pi\pi^*$ excited state decays the twisted intermediate rises, suggesting the formation of the twisted intermediate directly from the $^1\pi\pi^*$ excited state. The overlapping hot ground state and ground state bleach make it impossible to directly probe the formation timescale for the photohydrate to concretely determine a mechanism. However, kinetic isotope experiments by Mathew *et al.* suggest the decay of the twisted intermediate is driven by the hydration of uracil.⁶⁶ This evidence, along with the relatively long lifetime (compared to the excited state) and significant yield of the twisted form, strongly support the mechanism outlined in Figure 4. This mechanism is consistent with gas-phase calculations by Park *et al.* for the photohydration of uracil, which found the photoexcited $^1\pi\pi^*$

state rapidly decays to the electronic ground state *via* a conical intersection that either funnels molecules back to the ground state minimum, or a stable high-energy intermediate with a geometrically distorted half-chair puckered pyrimidine ring structure.⁶³ This key twisted intermediate is proposed by Park *et al.* to mediate photohydration with surrounding water molecules. In the last year, this intermediate has been invoked by Obara *et al.*⁶⁵ in time-resolved IR studies as the origin of a photoinduced absorption that persists for several picoseconds in thymine, uracil and their respective nucleosides, after electronic relaxation from the photoexcited $^1\pi\pi^*$ state.

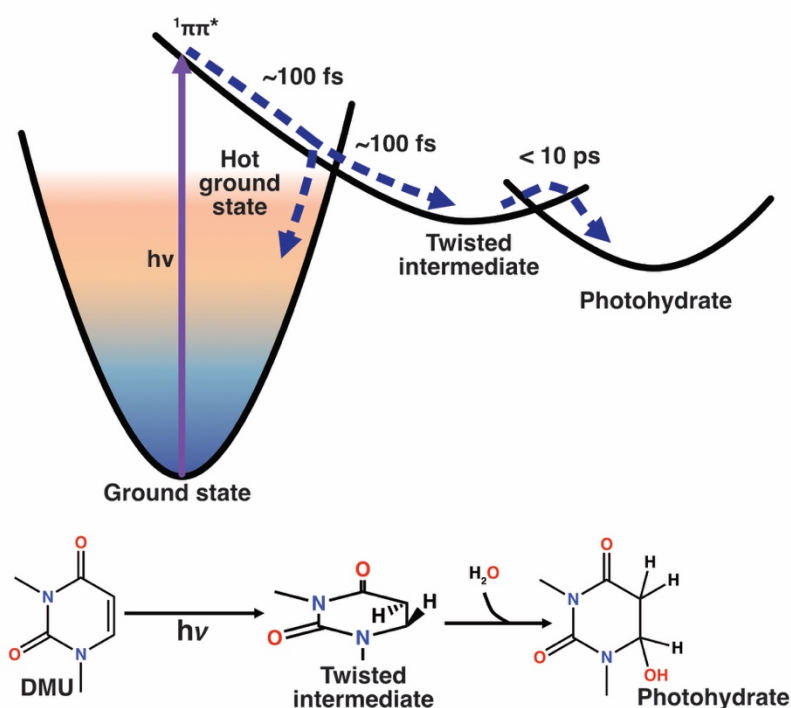


Figure 4 Reaction mechanism for the UV-induced photohydration of DMU. 4.6 eV irradiation excites DMU to the $^1\pi\pi^*$ photoexcited state. This population decays rapidly through a conical intersection with the ground state on a 100 fs timescale. As the hot ground state cools over < 10 ps, a small portion of molecules undergoes twisting with possible subsequent photohydration.

To fully capture the photochemical dynamics of photoexcited molecules in solution, ultrafast electronic spectroscopy requires probe continuum pulses spanning hundreds of nanometers due to the large variety of electronic signature associated with photochemical intermediates and reaction products, and the inherently broad line shapes of electronic transitions. A significant challenge that has faced the field has been how to produce a white light supercontinuum that spans multiple octaves from the NIR to the UV, with uniform spectral density throughout, and that can be used to measure a transient absorption spectrum of a solution phase sample on a shot-to-shot basis.

Hitherto, the best available solution has been to generate supercontinua from multiple different sources and acquire data sequentially, with laborious post-processing required to stitch these data together. Using the latest non-linear optical tools, we have addressed this deficiency by generating a soliton supercontinuum in gas-filled hollow-capillary fibers spanning 1.2–5.5 eV (220–1000 nm). We demonstrate this applied to solution phase transient absorption spectroscopy in the 2.1–5.5 eV region (with the NIR detection limited by current detection) using both Ti:Sapphire and Yb:KGW femtosecond amplifiers as the driving force for soliton generation.

The capabilities of this new spectrometer are demonstrated in two different systems, where a multi-octave probe spanning into the deep-UV provides clear and new photophysical and photochemical insights. The continuous non-resonant two-photon absorption spectrum of water has been measured between 6.8–10.2 eV, revealing information about broad overlapping electronic transitions. For photochemistry, we have shown direct detection for the formation of a photohydration product of DMU, which because of the removal of conjugation only absorbs at > 5.5 eV. DMU is a derivative of the RNA base uracil and a well-known actinometer, and our

investigation strongly suggests that the photoproduct is formed on the ground electronic state, and not directly from the photoexcited $^1\pi\pi^*$ state.

This instrument has the potential to outperform supercontinua generated in bulk media by offering significantly larger spectral bandwidths and higher photon flux. This advance opens the door to exploring previously inaccessible spectral regions in solution phase TA measurements, particularly the deep UV, where biomolecules like DNA and natural amino acids have electronic absorptions corresponding to key photoproducts or reaction intermediates. Further, we have demonstrated these findings using two different laser platforms, both of which are commonly used in the ultrafast community. We expect that soliton-generated supercontinuum will also find utility in multidimensional ultrafast spectroscopies where extension by a multioctave laser source could reveal the electronic couplings between many spectrally distinct electronic states and significantly enhance the associated time-resolution when combined with pulse shaping techniques.^{76,77}

Acknowledgements

JLGC and SEB acknowledge support from the Research Corporation for Science Advancement (CS-SEED-2022-003) and the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists, Office of Science Graduate Student Research (SCGSR) program, administered by the Oak Ridge Institute for Science and Education under contract number DE-SC0014664. SEB and TAAO acknowledge collaborative grants 2541785 from NSF MPS/CHE and UKRI2573 from EPSRC, respectively. TAAO acknowledges the Leverhulme Trust for the award of an International Fellowship (IF2024021) and the Royal Society for a University Research Fellowship (UF1402310, URF\R\201007). M.B. and R.F. were supported by the Department of Energy Office of Basic Energy Science, Facilities Division. J.C.T. is supported by

a Chair in Emerging Technology from the Royal Academy of Engineering. C.B. acknowledges support from the Royal Academy of Engineering through Research Fellowship No. RF/202122/21/133. The authors also acknowledge the contributions of Rose Wilson, who performed non-linear optical dynamical simulations to guide the initial stages of the project, and Yaguo Tang, who helped constructing the Yb:KGW-based transient absorption setup at SLAC.

Supporting Information Available: Comprehensive details of optical layout for soliton generation with Ti:Sapphire and Yb:KGW laser systems; simulations for tuning solution fission at fiber output; TA data from Yb:KGW pumped soliton; prism spectrograph implementation and calibration procedure; comparison of solvated electron signal from two different apparatus; sample details; data processing details.

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